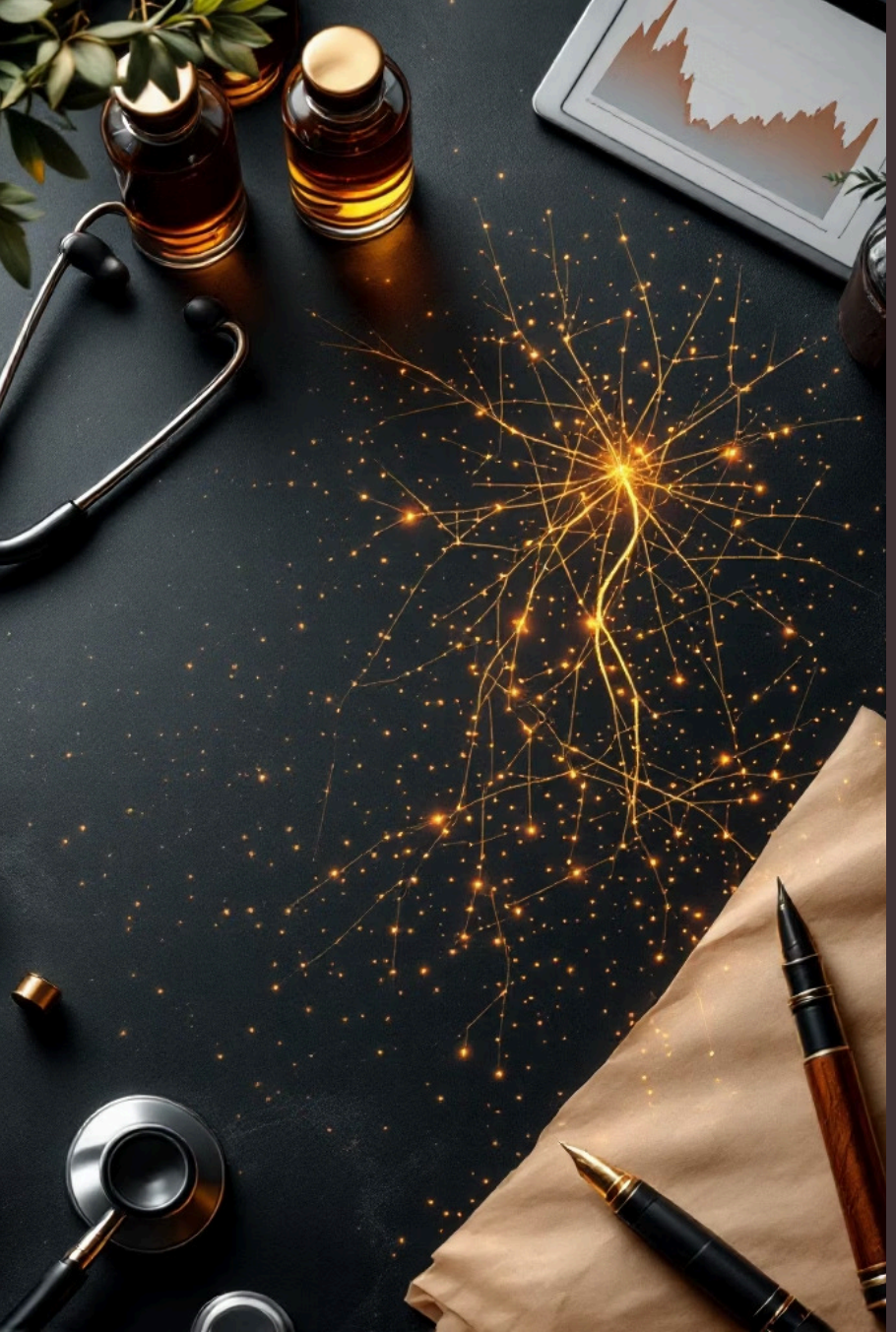

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Diabetes Risk Insight Engine

Scalable Predictive Analytics using **PySpark & Random Forest**

THE CHALLENGE

The Problem: Messy Data, Rising Risk

The Clinical Crisis

Diabetes affects over **537 million adults** globally. Early detection is critical — yet most healthcare systems lack scalable, automated risk stratification tools.

The Data Reality

Real-world clinical datasets are noisy, incomplete, and riddled with biologically impossible values. Traditional single-machine tools cannot handle this at scale.

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DATA SOURCE

Pima Indians Diabetes Dataset

A **gold-standard clinical benchmark** from the UCI Machine Learning Repository — widely used for diabetes prediction research.

768 Records

Female patients, Pima heritage, aged 21+

8 Parameters

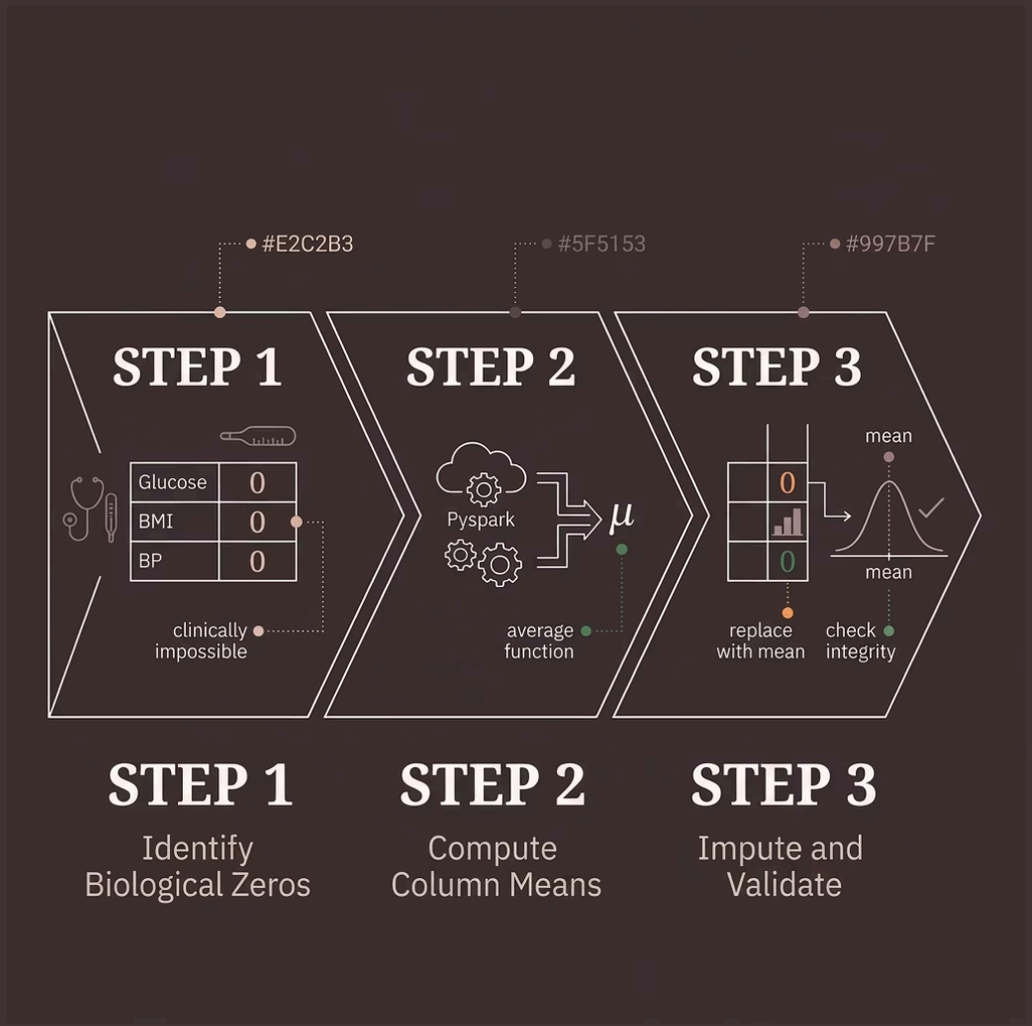
Glucose, BMI, Blood Pressure, Age & more

Binary Target

Positive (1) or Negative (0) diagnosis

DATA ENGINEERING

Cleaning "Biological Zeros"



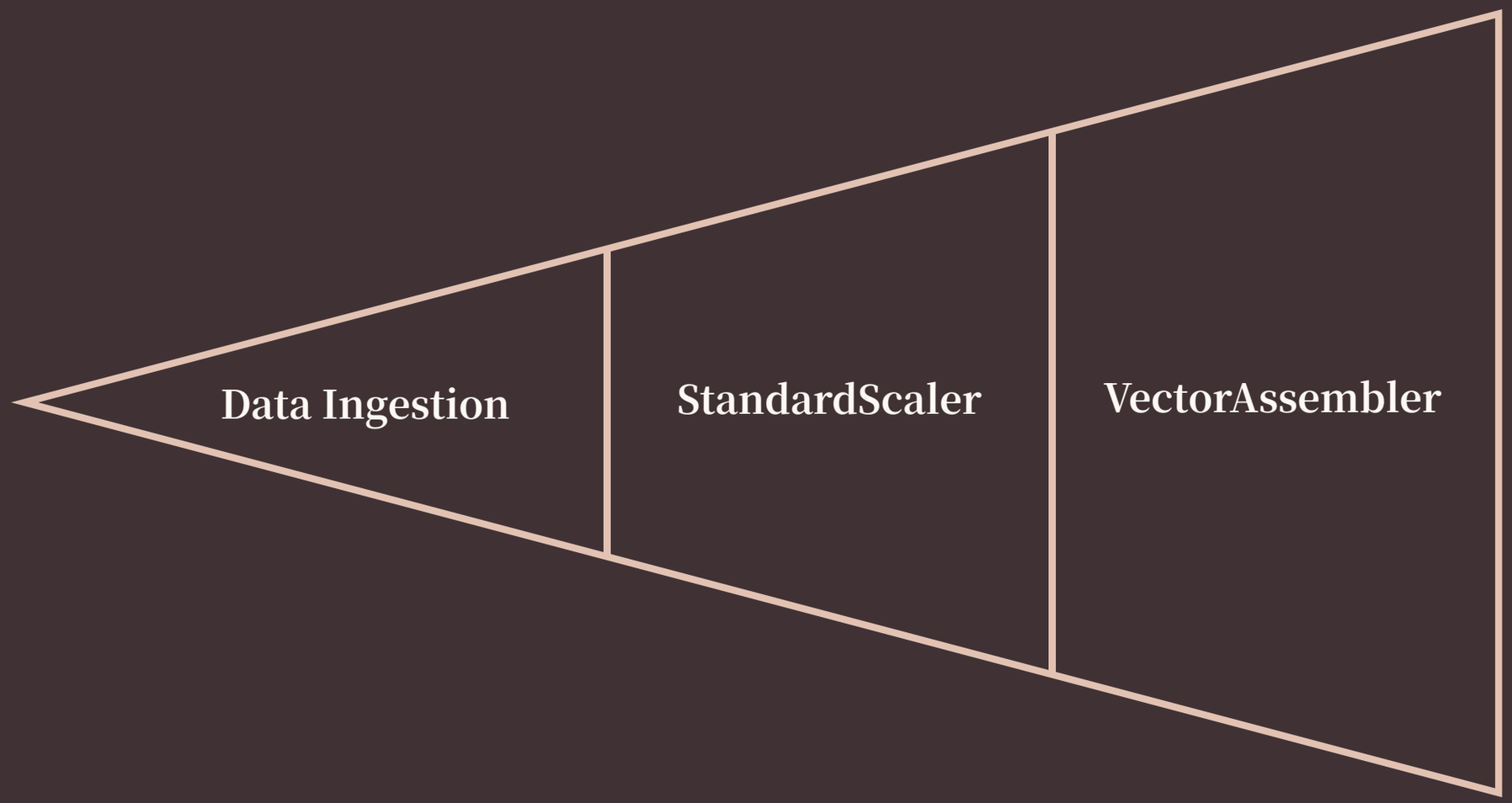
Why Mean Imputation in Spark?

Zeros in Glucose, BMI, and Blood Pressure are data-entry artifacts — not real measurements. Spark's distributed aggregation computes column means at scale, replacing outliers **without dropping rows** or compromising dataset integrity.

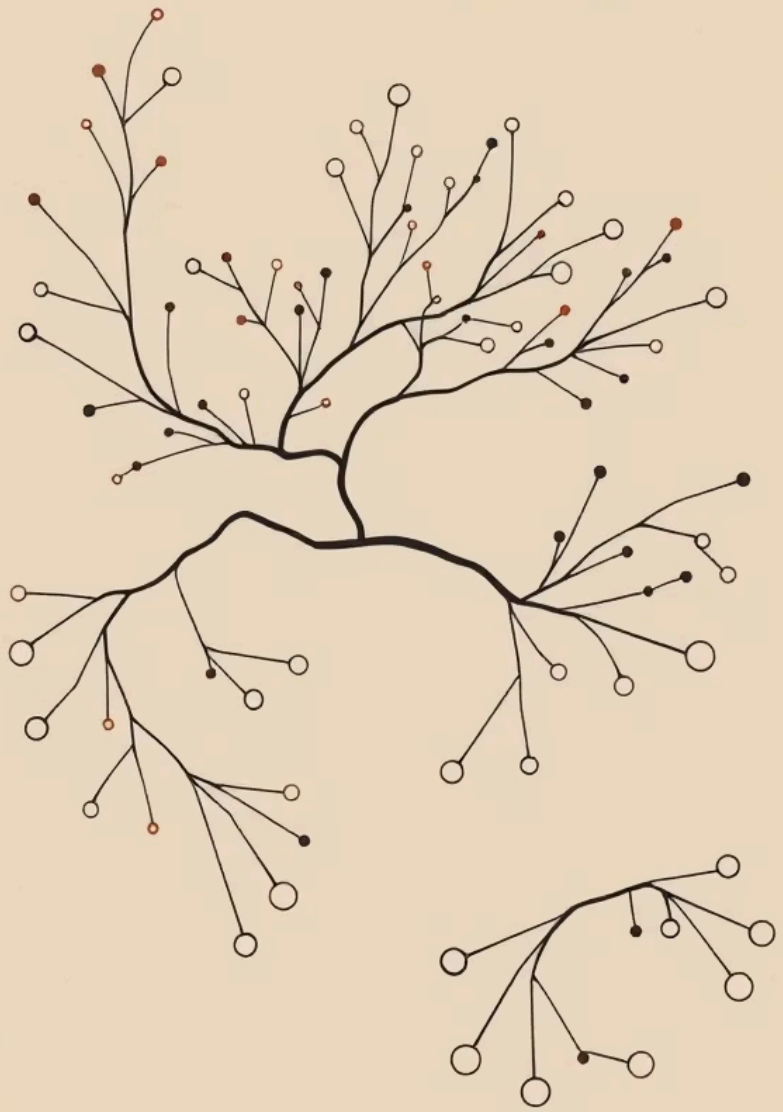
> Preserving all 768 records was critical for maintaining statistical power in a small clinical dataset.

PIPELINE ARCHITECTURE

The Backend: process.py



Standardisation is essential — features like Age (21–81) and Glucose (0–199) operate on vastly different scales. Without scaling, high-magnitude features would dominate the model unfairly.



MACHINE LEARNING

The Brain: Random Forest Classifier

Ensemble of 100 Trees

Each tree votes independently on the outcome. Aggregating predictions reduces variance and prevents overfitting — critical for clinical reliability.

Built-in Feature Importance

Random Forest natively ranks predictors by contribution, enabling clinical interpretability — Glucose emerges as the dominant signal.

Robust to Class Imbalance

With ~35% positive cases, the ensemble approach maintains sensitivity without sacrificing specificity on unseen data.

RESULTS

Model Performance on Unseen Data

0.77

F1-Score

Harmonic mean of precision and recall — balanced clinical utility

0.79

Precision

Of predicted positives, 79% were true diabetes cases

0.78

Recall

78% of actual diabetes cases correctly identified

These metrics confirm **reliable generalisation** on held-out clinical data — a strong foundation for real-world deployment.

THE DASHBOARD

The UI: Streamlit + Plotly

Real-Time Clinical Insight

The Streamlit dashboard surfaces model interpretability for non-technical stakeholders — clinicians and researchers can explore predictions without writing code.

🥇 Glucose

#1 predictor — highest feature importance score

🥈 BMI

Second strongest signal for diabetes risk

🥉 Age

Consistent contributor across all trees

CONCLUSION

A Scalable Foundation for Healthcare AI

🏗️ Scalable

PySpark architecture handles millions of patient records — ready for production-grade clinical pipelines.

🔬 Interpretable

Feature importance and dashboard visualisation ensure clinical trust and regulatory readiness.

📦 Open Source

Full codebase available on GitHub — reproducible, extensible, and community-ready.

🔗 **GitHub:** CLICK HERE— Explore the full pipeline, from data ingestion to dashboard deployment.

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Thank U.....